PROJECT AND TEAM INFORMATION

Project Title

BudgetBee – Small savings, bigger goals

Student / Team Information

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Course: DSA with C/C++ | B.Tech CSE (AI & ML) | Semester III

PROPOSAL DESCRIPTION

Motivation

Managing personal money becomes difficult when small expenses, bills and regular payments are recorded separately. Grocery shopping, electricity bills, EMIs, subscriptions and daily spending can easily get mixed up.

BudgetBee is planned to keep these records together. A user can enter income and expenses, divide them into categories and set a monthly limit. Regular payments such as rent, electricity, subscriptions and EMIs can also be kept in the same record.

The system is useful because it does not only record expenses. It also helps the user understand where money is being spent and how much is available after regular commitments.

The main idea is to make everyday expense tracking simple and practical.

State of the Art / Current Solution

At present, people commonly use notebooks, spreadsheets, banking applications or separate budgeting apps for tracking money. These options work for basic record keeping, but checking all expenses together can still take extra effort.

For example, an EMI may be remembered separately from grocery spending, while subscriptions may continue without being noticed. A person may also find it difficult to compare spending across different categories.

BudgetBee tries to bring different types of payments into one simple system. The system will use data structures to make storing, searching, sorting and grouping transactions easier.

This approach also gives practical use to concepts such as linked lists, hashing, trees, searching and sorting.

PROJECT GOALS AND MILESTONES

Project Goals

- Record income and expenses with useful categories.
- Set monthly budgets and give alerts near spending limits.
- Track EMIs, rent, electricity bills and subscriptions.
- Search transactions using date, amount or category.
- Sort expenses according to user's requirement.
- Detect unusually high expenses.
- Track savings targets and upcoming payments.
- Add CSV export for stored financial records.

Milestones

1. Design transaction and category structure.
2. Implement required data structures in C++.
3. Add income, expenses, budgets and recurring payments.
4. Implement searching and sorting.
5. Add spending analysis and alerts.
6. Test complete system with different inputs.

Project Approach

BudgetBee will be developed as a standalone C++ system in Visual Studio Code. It will not depend on a website or mobile application.

The transaction history will be maintained using a linked list. A hash map will support quicker category-wise lookup, while a Binary Search Tree will be used where organised searching is needed. Sorting algorithms will arrange records by date, amount or category.

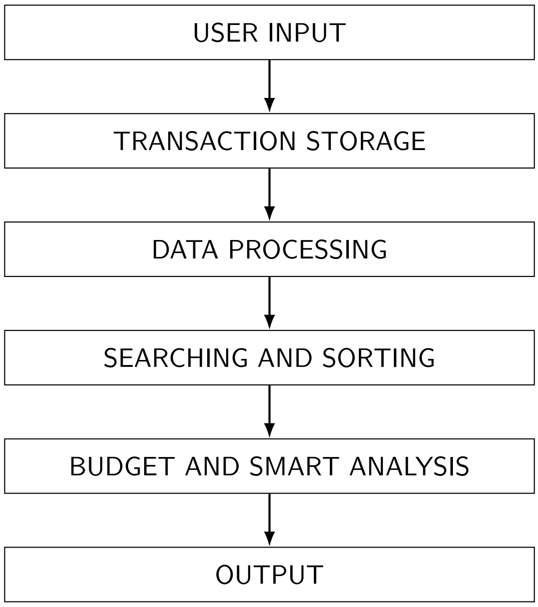
File handling will be used to save and load the user's records. The project will be divided into modules such as transaction handling, budgeting, payment tracking, searching, sorting and smart analysis.

Each module will be tested separately before combining the complete system. The final system will provide a simple menu-based interface through which the user can enter, search, analyse and manage financial records.

The project will focus on applying DSA concepts to a practical real-life problem rather than only demonstrating them theoretically.

SYSTEM ARCHITECTURE AND OUTCOME

System Architecture



Project Outcome / Deliverables

The expected result is a working C++ based personal finance system that keeps the user's income, expenses, budgets and regular payments together.

The main deliverables are:

- Income and expense recording.
- Category-wise monthly budgets and alerts.
- EMI, bill and subscription tracking.
- Transaction searching and sorting.
- Upcoming payment information.
- Unusual expense detection.
- Savings goal tracking.
- CSV export of financial records.

The completed system should help the user understand spending patterns and make better day-to-day budgeting decisions.

It will also demonstrate practical implementation of linked lists, hash maps, Binary Search Trees, searching, sorting and file handling in C++.

ASSUMPTIONS AND REFERENCES

Assumptions

- User enters correct amounts, dates and categories.
- System mainly for personal expense tracking.
- No direct bank account connection.
- Records initially stored locally.
- Analysis quality depends on entered records.
- Smart suggestions are basic rule-based.
- Intended for academic demonstration and personal use.

References

1. Mark Allen Weiss, *Data Structures and Algorithm Analysis in C++*, Pearson.
2. Bjarne Stroustrup, *Programming: Principles and Practice Using C++*, Addison-Wesley.
3. E. Balagurusamy, *Object Oriented Programming with C++*, McGraw Hill.
4. C++ Standard Library documentation used during implementation.
5. Course notes and classroom material for Data Structures and Algorithms with C/C++.